

Substring Order II

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Time limit: 1.00 s **Memory limit:** 512 MB

You are given a string of length n . If all of its substrings (not necessarily distinct) are ordered lexicographically, what is the k th smallest of them?

Input

The first input line has a string of length n that consists of characters a–z.

The second input line has an integer k .

Output

Print the k th smallest substring in lexicographical order.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq k \leq \frac{n(n+1)}{2}$